

Roll No.:

END SEMESTER EXAMINATION MAY 2025

Name of the Program: B. Tech (CSE)

Semester: 8th

Name of the Course: Agile Software Engineering

Course Code: TCS855

Time: 3 Hours

Maximum Marks: 100

Note:

- i. **All questions are compulsory.**
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Explain the core principles and features that define Agile Software Engineering. How are these applied in real-world software development?	CO1
(b)	Describe the life cycle of a Scrum project. How does it ensure continuous delivery and customer satisfaction?	CO1
(c)	Discuss how Pair Programming supports Agile principles such as collaboration, continuous feedback, and high code quality. Provide examples.	CO4
Q2		
(a)	A software development team is about to begin work on a new e-commerce application using Scrum. During the first sprint planning meeting, the Product Owner brings a backlog of user stories, but the team seems confused about their roles and responsibilities. The Scrum Master is also new and unsure how to facilitate the process effectively. 1. Identify and explain the roles in Scrum. 2. How should the Scrum Master handle this situation to ensure an effective sprint planning session?	CO2
(b)	Describe the major phases of an Agile project. Discuss how requirement gathering, design, development, testing, and deployment are handled in Agile project phases.	CO3
(c)	Explain the process of sprint planning with respect to estimation. How are story points used to determine the sprint backlog?	CO2
Q3		
(a)	How do Agile design principles support simplicity and flexibility in software systems? Explain with UML-based examples.	CO3
(b)	How does refactoring contribute to the long-term success of an Agile project? Discuss the risks of ignoring refactoring and the impact on the software's evolution.	CO3
(c)	A newly-formed Agile development team is assigned to build a customer relationship management (CRM) system for a small business. The team is unfamiliar with Test-Driven Development (TDD) and prefers to write code first and test later. However, the team faces challenges like frequent bugs, slow debugging, and a lack of confidence in their releases.	CO4

	How can Test-Driven Development (TDD) address the challenges faced by this Agile team? Explain the TDD cycle in detail and describe how it would improve code quality.	
Q4		
(a)	In Agile, testing is often integrated throughout the entire project lifecycle. Discuss how the Agile lifecycle impacts the role of testing in comparison to traditional waterfall models. What are the key benefits of this integrated approach?	CO5
(b)	Describe the process of conducting exploratory testing. How do testers document their findings, and how do they collaborate with the development team to ensure defects are resolved efficiently?	CO5
(c)	An Agile team is working on an e-commerce platform that regularly receives new feature updates, such as improved payment gateway integrations and new product categories. With each sprint, the development team implements new functionality, but there are concerns about the impact on existing features like user login, product search, and order processing. The product owner has requested that the team ensures no existing functionality is broken after the new features are deployed. 1. How would you manage regression testing in this Agile environment with frequent feature updates? 2. Discuss the role of automation in regression testing for this project.	CO5
Q5		
(a)	A tech start-up, which has been developing mobile applications using traditional waterfall methodologies, decides to shift to Agile practices in response to increasing competition and the need for faster delivery of features. The company's small team is enthusiastic about the change, but there is concern about whether Agile practices will scale as the company grows. The company has limited experience with Agile frameworks and needs guidance on how to implement it effectively. 1. What are the advantages of Agile adoption for a tech start-up looking to increase speed and adaptability? 2. How can the company manage the transition to Agile, given its small team size and limited experience with Agile frameworks?	CO6
(b)	A mobile app development team has been tasked with building a new application for a client, which will run on the cloud. The client requires the app to handle dynamic user data and scale according to demand. The team uses Agile methodologies and has decided to utilize cloud-based infrastructure for hosting the app and managing backend services. However, the team is unsure how to align their Agile practices with cloud services, especially when dealing with frequent updates, real-time data processing, and ensuring security in the cloud. 1. Discuss how cloud infrastructure can support Agile practices such as iterative development and frequent releases. 2. What are the potential risks when integrating cloud infrastructure with an Agile project, especially in mobile app development?	CO6
(c)	Identify the key risks associated with Agile projects and discuss strategies to mitigate these risks. How do teams address risks such as changing requirements, team turnover, and stakeholder misalignment?	CO6